

BhoomiSetu — Officer Roles: Detailed Workflow & Unique Value-Add

Verdict on Your 7 Officer Roles

Keep them as-is — don't change the list. Here's why each one maps directly to a real gap named in the problem statement:

Your Officer	Real-world equivalent	PS pain point it solves
Land Record Officer	Talathi / Tehsildar (RoR, 7/12, mutation)	"Cadastral maps, Record of Rights... managed independently"
Registration Officer	Sub-Registrar (IGR department)	"Registration records"
Planning Officer	Town Planning / Master Plan authority	"Land use information, Master Plan, Building Permission"
Tax Officer	Revenue/Municipal tax dept	"Property taxation records"
Restriction Officer	Collector's office (ceiling, forest, government land flags)	"Restrictions"
Encumbrance Officer	Sub-Registrar's encumbrance wing	"Utility infrastructure / other land-related databases" (loans, mortgages, liens)
Dispute Officer	Tehsildar's Revenue Court function	Not explicitly named in PS, but essential — this is your operational glue connecting citizen complaints across all the above

This 7-way split is actually a strong answer to the PS's core complaint: *"land governance involves multiple institutions maintaining information in fragmented, disconnected systems."* You've modeled each fragment as its own role instead of pretending it's all one department — which is more realistic and more defensible to evaluators than a single generic "Officer" role.

One optional addition to consider (not required): A **Survey Officer** role for physical measurement/GIS corrections — currently this might sit inside "Land Record Officer," which is fine for a prototype, but call this out explicitly if an evaluator asks "who updates the map geometry itself?"

Detailed Workflow — What Each Officer Actually Does

1. Land Record Officer (landrecords.officer@)

Real-world job: Maintains RoR (7/12/8A), approves mutations, verifies ownership.

Workflow:

1. Login → Dashboard shows queue: pending mutation requests, name/area correction requests (from your OCR verification flow).
2. Opens a request → sees the AI-generated structured message + attached proof + OCR match score.
3. Cross-checks against the current RoR entry in the system.
4. Actions available: **Approve** (updates RoR, triggers new parcel version + audit entry) / **Reject with reason** / **Send back for more info**.
5. On approval, the citizen's parcel status flips from "Pending Verification" → "Registered" (this is the exact gate you built for Raise Complaint).

Unique BhoomiSetu solution: Instead of the officer manually re-typing/re-checking data from a scanned document, they see the **OCR-extracted fields side-by-side with citizen-typed fields and a match %** — cutting manual verification time from a full site/record check to a quick confirm-or-flag decision.

2. Registration Officer (registration.officer@)

Real-world job: Handles sale deed registration, links registered documents to the RoR update chain.

Workflow:

1. Queue: pending registration-linked requests (e.g., "my sale deed isn't reflected in the record").
2. Verifies the registered document reference (registration number, date, parties) against uploaded proof.
3. Confirms → sends a linking signal to the Land Record Officer's queue so mutation can proceed (registration and mutation are legally separate steps — your system should model this handoff explicitly, not merge them).
4. Can flag suspected duplicate/conflicting registrations on the same survey number.

Unique BhoomiSetu solution: Automatic duplicate-registration flagging — when a new registration references a survey number that already has an active registered owner, the system flags it before the officer even opens the file, instead of relying on the officer to remember/cross-check manually.

3. Planning Officer ((planning.officer@))

Real-world job: Manages land-use classification, Master Plan zoning, building permissions.

Workflow:

1. Queue: building permission requests, land-use change requests, zoning queries.
2. Checks the parcel against the Master Plan layer (GIS zoning overlay) — is the requested use allowed in this zone?
3. Approves/rejects with a zoning-compliance note; if approved, updates the parcel's land-use attribute.

Unique BhoomiSetu solution: Because parcels are geometry-linked (not just text records), the officer sees an **automatic zoning-conflict check** — the system visually overlays the parcel on the zoning map and highlights (in red) if the requested use doesn't match the zone, before manual review even starts.

4. Dispute Officer ((dispute.officer@))

Real-world job: Tehsildar-equivalent revenue court function — hears boundary disputes, ownership conflicts, encroachment complaints.

Workflow:

1. Queue: complaints classified as "Dispute Filing" (via your AI classification/"Describe My Issue" flow).
2. Reviews the structured message, attached proof, and — if physical verification was triggered — the field verifier's GPS-tagged photo evidence.
3. Can escalate to Collector-level review for unresolved/high-value disputes (mirrors the real SDO → Collector → Commissioner appeal chain).
4. Final ruling recorded with reasoning, visible to the citizen via status tracking.

Unique BhoomiSetu solution: The **evidence chain is pre-assembled before the officer even opens the case** — citizen's complaint, OCR-verified document, field verifier's geo-tagged photos, and any prior rejected-claim history on that survey number, all in one view — instead of the officer chasing paperwork across departments.

5. Tax Officer ((tax.officer@))

Real-world job: Manages land revenue/property tax assessment and payment records.

Workflow:

1. Queue: tax dispute requests ("my assessment is wrong"), reassessment requests after mutation.

2. Checks parcel area/classification (agricultural vs non-agricultural — tax rates differ) against the current RoR.
3. Updates assessment, generates/updates the tax record tied to the parcel.

Unique BhoomiSetu solution: Tax reassessment is **auto-triggered** when the Land Record Officer approves a mutation with a changed area/classification, instead of requiring the citizen to separately go file a tax update request — one verified change propagates across departments.

6. Restriction Officer (`restriction.officer@`)

Real-world job: Manages statutory restrictions — ceiling-act land, forest land, government/gairan land, prohibited-transfer flags.

Workflow:

1. Queue: requests to add/remove/verify a restriction flag on a parcel (e.g., a citizen disputing that their land is wrongly flagged as restricted).
2. Cross-checks against the restriction source record (forest department notification, ceiling act order, etc. — as an uploaded reference document).
3. Approves/rejects the flag change; this directly affects whether that parcel is transferable/mortgageable elsewhere in the system.

Unique BhoomiSetu solution: Restriction flags are **enforced at the data layer**, not just displayed — if a parcel has an active restriction, the system automatically blocks/warns on registration or encumbrance actions elsewhere in the platform, so a restricted parcel can't accidentally get registered or mortgaged through a different officer's queue.

7. Encumbrance Officer (`encumbrance.officer@`)

Real-world job: Maintains records of loans, mortgages, and liens registered against a parcel (encumbrance certificate).

Workflow:

1. Queue: encumbrance certificate requests, new-loan/mortgage registration entries from banks (e.g., Kisan Credit Card loans against land).
2. Verifies the parcel is not already over-leveraged or under an active dispute/restriction before approving a new encumbrance entry.
3. Issues the encumbrance certificate (PDF) or updates the encumbrance ledger for that parcel.

Unique BhoomiSetu solution: Because encumbrance, dispute, and restriction data live on the **same unified parcel record** (not separate silos), the officer automatically sees if a parcel under

dispute or restriction is being used for a new loan application — preventing exactly the kind of fraud (mortgaging disputed/restricted land) that fragmented systems allow today.

The One Line That Ties All 7 Together (for your demo)

"Each officer only sees their department's slice of work — but every action they take writes back to **one shared parcel record**, so a mutation approval automatically updates tax liability, a restriction flag automatically blocks encumbrance registration, and a dispute automatically pauses any pending registration on that same survey number. That cross-department consistency is the actual problem statement — not any single officer's dashboard."**